Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/02/2019
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jimq@umd.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
Organization Name:	University of Maryland		
PI Address 1:	Fire Protection Engineering		
PI Address 2:	3104G Jm Patterson		
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	10/08/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 11/7/19) NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	80NSSC20M0119 ; NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station (ISS). The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows: 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	Two journal papers were published in 2019, one was accepted, and a fourth was submitted. One M.S. student defended his thesis on BRE and graduated. Round I of BRE testing on the ISS was performed in 2019 and is now complete. A total of 63 BRE flames were observed on the 25 mm burner during eight test days. The results are highly significant. Many flames were observed to burn steadily, which was not expected by most microgravity combustion experts. We have been analyzing the data for all 63 ISS tests, including the radiometers, heat flux gauges, and calorimeter. We developed a flammability map for steady flames. We obtained heats of gasification for all 29 steady tests using the calorimeter. On every ISS test the heat flux sensors in the burner surface experience discontinuities. These are upward jumps in sensor output, from typically 3-6 kW/m2, typically about 60 s after ignition. These jumps are attributed to water condensation and evaporation. We performed normal gravity experiments to optimize the pre-flow sequence for Advanced Combustion via Microgravity Experiments (ACME) ISS tests. This is required owing to the large plenum in the BRE burners. In normal gravity we successfully emulated methanol pool fires of methanol, acetone, and ethanol. For this we matched heat of combustion, smoke point, heat of gasification, and flame height. We began emulating several plastics. We continued to analyze normal-gravity tests with propane. This included fitting flame height correlations and analyzing the theoretical convective heat flux. The heat flux readings and the temperatures of the heat flux sensors seem to be reversed from what is expected. We investigated if these wires could be crossed or the data reversed somehow. We performed paint degradation tests to study the heat treatment effect on the calibration constant. We repainted one of the heat flux sensors to investigate possible paint degradation. Any degradation has only a minor effect on the measurements. We calibrated the top-plate calorimeters for the various burners in normal gravity and microgravity. This requires a detailed heat flux analysis and yields a set of calorimeter constants for each burner. We optimized the calorimeter calibration for the 25 mm ISS burner. This was confirmed by the center heat flux gauge and the radiant fraction. We explored extensions to the Baum ellipsoidal model. This is now able to predict the transient shapes of the ISS flames. We worked on the radiation analysis of the ISS tests with three different methods. These were Mean beam length; Thin hemispherical flame; and Thin ellipsoidal flame with considering the flame standoff distance. These analysis help estimate the radiant fraction and the radiative flux to the burner surface as a function of radius. Alex Snegirev and his team continued to perform Fluent simulations of the BRE flames in microgravity. Work continued in finalizing the ISS test matrix for Round II and a possible bonus round.

Bibliography Type:	Description: (Last Updated: 10/08/2025)
Articles in Peer-reviewed Journals	Markan A, Baum HR, Sunderland PB, Quintiere JG, de Ris JL. "Transient ellipsoidal combustion model for a porous burner in microgravity." Combustion and Flame. 2020 Feb;212:93-106. https://doi.org/10.1016/j.combustflame.2019.09.030 , Feb-2020
Articles in Peer-reviewed Journals	Vermina Plathner F, Quintiere JG, van Hees P. "Analysis of extinction and sustained ignition." Fire Safety Journal. 2019 Apr;105:51–61. https://doi.org/10.1016/j.firesaf.2019.02.003 , Apr-2019
Articles in Peer-reviewed Journals	Markan A, Sunderland PB, Quintiere JG, de Ris JL, Baum HR. "Measuring heat flux to a porous burner in microgravity." Proceedings of the Combustion Institute. 2019 Jan;37(3):4137-44. https://doi.org/10.1016/j.proci.2018.05.006 , Jan-2019
Articles in Peer-reviewed Journals	Kuznetsov EA, Snegirev AY, Markus ES. "Radiative extinction of laminar diffusion flame above the flat porous burner in microgravity: A computational study." Combustion, Explosion, and Shock Waves. 2020 Aug;56(4):394-411. https://doi.org/10.1134/S0010508220040036 , Aug-2020